

# Supporting Information

## Coordinated Vibronic Light-Harvesting and Polaritonic Interface for Symbiotic Quantum Agrivoltaics

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June 25, 2026

**Associated Manuscript:** This Supporting Information accompanies the manuscript “Coordinated Vibronic Light-Harvesting and Polaritonic Interface for Symbiotic Quantum Agrivoltaics” submitted to *Nature Energy*. Equations, figures, and tables are prefixed with “S” (e.g., Eq. S1, Fig. S1, Table S1). References to the main manuscript are indicated by unadorned equation/figure numbers (e.g., Eq. 1, Fig. 1).

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## S1 8-Site FMO Hamiltonian and Full Coupling Matrix

### S1.1 Hamiltonian Formulation

The 8-site Fenna–Matthews–Olson (FMO) Hamiltonian used throughout this work extends the standard 7-site model [?] by including the eighth bacteriochlorophyll *a* molecule that bridges the chlorosome baseplate to the reaction center [?]. The full electronic Hamiltonian in the site basis is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (\text{S1})$$

where  $\varepsilon_n$  are the site energies (referenced to the lowest-energy site),  $J_{nm}$  are the inter-site electronic couplings, and  $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$  is the irreversible reaction center trapping rate applied to Sites 3 and 4. The anti-Hermitian term  $-i\Gamma_{\text{RC}}$  models irreversible charge separation at the reaction center; the Hamiltonian is non-Hermitian only on these diagonal trapping terms.

### S1.2 Site Energies

Table S1 lists the site energies used in all production simulations. Energies are given relative to the lowest-energy site (Site 3 at  $12\,210 \text{ cm}^{-1}$ ) and are expressed in  $\text{cm}^{-1}$ .

Table S1: **8-site FMO site energies** (in  $\text{cm}^{-1}$ , referenced to Site 3).

| Parameter                | Site 1 | Site 2 | Site 3 | Site 4 | Site 5 | Site 6 | Site 7 | Site 8 |
|--------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| Absolute $\varepsilon_n$ | 12410  | 12530  | 12210  | 12320  | 12480  | 12630  | 12440  | 12410  |
| Relative $\varepsilon_n$ | 280    | 420    | 0      | 110    | 270    | 500    | 310    | 200    |

The absolute site energies are taken from the Adolphs–Renger parameterization [?] for Sites 1–7. Site 8, which bridges the baseplate to the reaction center, is assigned an energy of  $12\,410 \text{ cm}^{-1}$  based on the eighth-bacteriochlorophyll characterization by Moix *et al.* [?].

### S1.3 Electronic Coupling Matrix

Table S2 presents the full  $8 \times 8$  electronic coupling matrix  $J_{nm}$  (in  $\text{cm}^{-1}$ ). The matrix is symmetric ( $J_{nm} = J_{mn}$ ), and all 28 unique coupling pairs are listed. Values for Sites 1–7 follow the Adolphs–Renger parameterization [?]; couplings involving Site 8 are taken from Ref. [?].

Table S2: **Full 8-site electronic coupling matrix**  $J_{nm}$  (in  $\text{cm}^{-1}$ ).

|               | Site 1 | Site 2 | Site 3 | Site 4 | Site 5 | Site 6 | Site 7 | Site 8 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|
| <b>Site 1</b> | 0      | -87.7  | 5.5    | -5.9   | 6.7    | -13.7  | -9.9   | 4.0    |
| <b>Site 2</b> |        | 0      | 30.8   | 8.2    | 0.7    | 11.8   | 4.3    | -2.1   |
| <b>Site 3</b> |        |        | 0      | -53.5  | -2.2   | -9.6   | 6.0    | 12.0   |
| <b>Site 4</b> |        |        |        | 0      | -70.7  | -17.0  | -63.3  | 5.5    |
| <b>Site 5</b> |        |        |        |        | 0      | 81.1   | -1.3   | 3.2    |
| <b>Site 6</b> |        |        |        |        |        | 0      | 39.7   | -8.7   |
| <b>Site 7</b> |        |        |        |        |        |        | 0      | -15.0  |
| <b>Site 8</b> |        |        |        |        |        |        |        | 0      |

## S1.4 Reaction Center Trapping

The irreversible trapping is implemented through a non-Hermitian term on Sites 3 and 4, which together form the primary funnel to the reaction center in the 8-site FMO model [?, ?]. The trapping rate  $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$  corresponds to a trapping time of  $\tau_{\text{RC}} \approx 6.7 \text{ ps}$ , consistent with ultrafast charge separation rates measured in reaction center complexes [?]. The forward transfer yield is defined as:

$$\Phi_{\text{FT}} = 2\Gamma_{\text{RC}} \int_0^{t_{\text{max}}} [P_3(t) + P_4(t)] dt, \quad (\text{S2})$$

with  $t_{\text{max}} = 1 \text{ ps}$  and  $P_j(t) = \rho_{jj}(t)$ . The factor of 2 accounts for the two trapping sites.

## S1.5 Hermiticity Validation

The non-Hermiticity is strictly confined to the diagonal entries of Sites 3 and 4. For all  $i \neq j$ ,  $H_{ij} = H_{ji}^*$  is satisfied within machine precision ( $1 \times 10^{-14}$ ). The Hamiltonian’s Hermiticity condition is verified for all 64 matrix elements at every time step of the simulation.

## S2 Convergence Analysis

### S2.1 Hierarchy Depth Convergence

We adopt  $L = 8$  as the production standard to ensure robust convergence across all disorder realizations. The numerical framework (PT-HOPS/SBD implementation, hierarchy equations) is directly inherited from the validation framework established in Paper I [?], where convergence was validated via MAE analysis on the same excitonic parameters.

Table S3 summarizes the mean absolute error (MAE) between successive  $L$  values and the relative deviation from the  $L = 9$  reference (computed for a subset of 10 disorder realizations).

Table S3: **Hierarchy depth convergence.** MAE between successive hierarchy depths for exciton populations, averaged over all 8 sites and 100 disorder realizations.

| $L$ | MAE ( $L \rightarrow L + 1$ ) | Rel. dev. from $L = 9$ | CPU time (min) |
|-----|-------------------------------|------------------------|----------------|
| 6   | $9.8 \times 10^{-10}$         | $1.2 \times 10^{-8}$   | 2.1            |
| 7   | $9.6 \times 10^{-10}$         | $1.1 \times 10^{-8}$   | 4.3            |
| 8   | $3.1 \times 10^{-11}$         | $4.2 \times 10^{-10}$  | 7.5            |

The MAE between  $L = 7$  and  $L = 8$  is  $3.1 \times 10^{-11}$ , which is below double-precision round-off for most observables. We adopt  $L = 8$  as the production standard to ensure robust convergence across all disorder realizations.

### S2.2 Matsubara Truncation

We test Matsubara term counts  $K = 1, 2, 3$  to assess the impact of high-temperature corrections. Table S4 shows the effect on the coherence lifetime  $\tau_c$  and the forward transfer yield  $\Phi_{\text{FT}}$ .

Table S4: **Matsubara truncation convergence.** Impact of  $K$  on key observables at 295 K.

| $K$ | $\tau_c$ (fs) | $\Phi_{\text{FT}}$ | MAE from $K = 3$     |
|-----|---------------|--------------------|----------------------|
| 1   | 406(30)       | 0.86(3)            | $3.4 \times 10^{-4}$ |
| 2   | 420(35)       | 0.89(3)            | $3.3 \times 10^{-5}$ |
| 3   | 422(36)       | 0.89(3)            | —                    |

The MAE between  $K = 2$  and  $K = 3$  is  $3.3 \times 10^{-5}$ , confirming that  $K = 2$  is sufficient for converged dynamics at room temperature. This is consistent with the expectation that at 295 K, the dynamics are dominated by the explicit vibronic modes rather than high-order Matsubara corrections to the Drude–Lorentz background.

### S2.3 Time Step Convergence

We verify time step convergence by comparing  $\Delta t = 0.2$  fs, 1.0 fs and 2.0 fs. Table S5 summarizes the results.

Table S5: **Time step convergence.** Maximum population deviation relative to the  $\Delta t = 0.2$  fs reference.

| $\Delta t$ (fs) | Max. population deviation |
|-----------------|---------------------------|
| 0.5             | — (reference)             |
| 1.0             | $4.3 \times 10^{-3}$      |
| 2.0             | $8.1 \times 10^{-3}$      |

We adopt  $\Delta t = 0.2$  fs as the production standard, consistent with Paper I [?].

### S2.4 Detailed Balance

The steady-state populations are compared with the analytical Boltzmann distribution to verify detailed balance. The maximum deviation across all 8 sites is 2.3 %, confirming that thermal equilibrium is correctly reproduced.

### S2.5 Physical Consistency Checks

All production simulations satisfy the following consistency checks:

- **Trace preservation:**  $\max_t |\text{Tr}[\rho(t)] - 1| < 1 \times 10^{-12}$
- **Positivity:**  $\min_i \lambda_i(t) > -1 \times 10^{-14}$  for all eigenvalues  $\lambda_i$  of  $\rho(t)$
- **Hermiticity:**  $\max_{i,j,t} |\rho_{ij}(t) - \rho_{ji}^*(t)| < 1 \times 10^{-14}$
- **Population realness:**  $\max_{i,t} |\text{Im}[\rho_{ii}(t)]| < 1 \times 10^{-14}$

## S3 Polaron Dressing Mechanism and Residual Reorganization Energy

### S3.1 Canonical (Lang–Firsov) Transformation

The polaron dressing mechanism that underlies the coherence protection reported in the main text can be rigorously understood through the canonical Lang–Firsov transformation [?, ?]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k} (\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \quad (\text{S3})$$

where  $g_{n,k}$  are the system-bath coupling constants for mode  $k$  at site  $n$ , and  $\hat{b}_{n,k}$  ( $\hat{b}_{n,k}^\dagger$ ) are the bath annihilation (creation) operators.

The unitary transformation  $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k} (\hat{b}_{n,k}^\dagger - \hat{b}_{n,k})\right]$  partially diagonalizes the system-bath coupling. After applying  $\hat{U}$ , the transformed Hamiltonian becomes:

$$\hat{H}' = \hat{U}^\dagger \hat{H} \hat{U} = \hat{H}'_S + \hat{H}_{\text{bath}} + \hat{H}_{\text{res}}, \quad (\text{S4})$$

where  $\hat{H}'_S$  contains renormalized excitonic couplings (the polaron-shifted energies) and  $\hat{H}_{\text{res}}$  represents the residual system-bath coupling after the transformation.

### S3.2 Residual Spectral Density

The key physical insight is that underdamped vibronic modes satisfying the resonance condition  $\Delta E_{nm} \approx \hbar\omega_v \pm J_{nm}$  become partially “dressed” by the canonical transformation: they are incorporated into  $\hat{H}'_S$  and no longer contribute to dephasing in the transformed frame. The residual spectral density is:

$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \quad (\text{S5})$$

where  $\mathcal{S}$  denotes the set of underdamped modes that are vibronically dressed.

### S3.3 Quantitative Effect

For the 12-mode Kleinekathöfer/Coker parameterization at 295 K, the residual reorganization energy under the two excitation scenarios is:

$$\lambda_{\text{res}}^{\text{broad}} = \lambda_{\text{DL}} + \sum_{k=1}^{12} S_k \omega_k \approx 53 \text{ cm}^{-1}, \quad (\text{S6})$$

$$\lambda_{\text{res}}^{\text{filt}} = \lambda_{\text{DL}} + \sum_{k \notin \mathcal{S}} S_k \omega_k \approx 40 \text{ cm}^{-1}. \quad (\text{S7})$$

The 20 % reduction in  $\lambda_{\text{res}}$  translates to a 50 % coherence lifetime extension (since  $\tau_c \propto 1/\lambda_{\text{res}}$  in the high-temperature limit). The dual-band filter at 750 nm and 820 nm selectively excites those vibronic resonances that are most strongly coupled to the energy-transfer pathway (primarily the  $180 \text{ cm}^{-1}$  Franck–Condon mode at Sites 3/4 and the  $740 \text{ cm}^{-1}$  ring deformation mode), maximizing the dressing effect.

### S3.4 Connection to PT-HOPS/SBD

In the PT-HOPS/SBD simulation framework, this protection is captured natively: the bath correlation function is decomposed into Drude–Lorentz and underdamped components, and the hierarchy equations propagate the system-bath entanglement without assuming weak coupling. The resulting trajectories exhibit the suppressed decoherence predicted by the vibronic-basis analysis when the pulse spectrum matches the underdamped resonances.

## S4 Dynamic Photo-Protection: Floquet Stark and OMIT Behavior

### S4.1 Floquet Stark Detuning

Under high solar flux ( $> 800 \text{ W m}^{-2}$ ), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift. The driving term in the Hamiltonian is:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (\text{S8})$$

with  $V_0 = 0.05 \text{ eV}$  and  $\omega_F = 1.2 \text{ THz}$ . At  $t = 0$  (zero flux), the cosine term evaluates to  $V_0$ , ensuring continuity with the undriven Hamiltonian. When the flux drops below threshold, the amplitude  $V_0$  decays to zero with a relaxation time  $\tau_{\text{relax}} = 10 \text{ ps}$ , preventing numerical singularities.

## S4.2 OMIT Transmission Modulation

The optomechanically induced transparency (OMIT) mechanism [?] creates destructive interference in the NPoM cavity that suppresses transmission at the resonant channels. The transmission coefficient is:

$$T_{\text{OMIT}}(I) = \begin{cases} T_0, & I = 0, \\ T_0 \left(1 + \frac{I}{I_{\text{sat}}}\right)^{-1}, & I > 0, \end{cases} \quad (\text{S9})$$

where  $I_{\text{sat}} = 800 \text{ W m}^{-2}$ . The piecewise definition at  $I = 0$  ensures continuity:  $T_{\text{OMIT}}(0) = T_0$ , corresponding to the undriven cavity transmission. At high flux ( $1000 \text{ W m}^{-2}$ ),  $T = T_0 \times (1 + 1000/800)^{-1} = 0.44 T_0$ , i.e., transmission drops to 44 % of baseline. The time-dependent flux follows 24 h solar cycles, and the OMIT response is instantaneous on agricultural timescales.

## S4.3 Night Mode (Zero Flux)

At zero flux (night mode), both photo-protection mechanisms return to their baseline states:

- Floquet Stark:  $V_0 \rightarrow 0$ , restoring the undriven OPV–FMO coupling.
- OMIT:  $T \rightarrow T_0$ , restoring full transmission.

No singularities or discontinuities arise because both models are explicitly defined for  $I = 0$ .

# S5 Socioeconomic Analysis: Cooperative Payback Model

## S5.1 Capital Expenditure and Cost Structure

The capital expenditure (CAPEX) for the quantum agrivoltaic system (Scenario A) comprises:

- **NPoM-coated graphene barrier:**  $120 \text{ m}^{-2}$  (including gold nanoparticle synthesis and deposition)
- **OPV module:**  $80 \text{ m}^{-2}$  (blade-coated semi-transparent polymer cells)
- **Installation and mounting:**  $50 \text{ m}^{-2}$
- **Electronics and IoT:**  $30 \text{ m}^{-2}$  (sensors, microcontroller, QKD module)
- **Contingency (15 %):**  $42 \text{ m}^{-2}$

Total CAPEX:  $322 \text{ m}^{-2}$ .

## S5.2 Cooperative Ownership Model

We model a cooperative of  $N$  smallholder farmers who collectively own and operate the quantum agrivoltaic system. The system size is assumed to scale with cooperative membership: a 5-member cooperative manages a  $500 \text{ m}^2$  installation.

The payback period is computed as:

$$P = \frac{C_{\text{total}}(1 - s)}{N \times (R_{\text{elec}} + R_{\text{water}} + R_{\text{crop}})}, \quad (\text{S10})$$

where  $C_{\text{total}} = 322 \text{ m}^{-2} \times A$  is the total CAPEX for area  $A$ ,  $s$  is the subsidy rate (fraction of CAPEX covered by carbon-linked subsidies),  $R_{\text{elec}}$  is annual electricity revenue,  $R_{\text{water}}$  is water savings value, and  $R_{\text{crop}}$  is crop yield improvement revenue.

### S5.3 Revenue Streams

- **Electricity:**  $180 \text{ kW h m}^{-2} \text{ yr}^{-1}$  at  $0.12 \text{ kW}^{-1} \text{ h} = 21.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Water savings:**  $1300 \text{ L m}^{-2} \text{ yr}^{-1}$  at  $0.002 \text{ L}^{-1} = 2.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Crop yield improvement:** 15 % increase at  $1.5 \text{ kg}^{-1} = 5.0 \text{ m}^{-2} \text{ yr}^{-1}$
- **Carbon credit (conservative):**  $12.4 \text{ kg CO}_2\text{e/m}^2/\text{yr}$  at  $10 \text{ USD t}^{-1} = 0.124 \text{ m}^{-2} \text{ yr}^{-1}$

Total annual revenue:  $29.3 \text{ m}^{-2} \text{ yr}^{-1}$ .

### S5.4 Payback Periods

Table S6 summarizes the payback period for various cooperative sizes and subsidy rates.

Table S6: **Payback period (years) as a function of cooperative size  $N$  and subsidy rate  $s$ .**

| $N$            | $s = 0 \%$ | $s = 20 \%$ | $s = 30 \%$ | $s = 50 \%$ |
|----------------|------------|-------------|-------------|-------------|
| 1 (individual) | 14.2       | 10.8        | 8.2         | 5.1         |
| 3              | 7.8        | 5.9         | 4.5         | 2.8         |
| 5              | 5.9        | 4.5         | <b>3.5</b>  | 2.1         |
| 10             | 4.2        | 3.2         | 2.4         | 1.5         |

For a 5-member cooperative with a 30 % carbon-linked subsidy, the payback period of 3.5 yr makes the quantum agrivoltaic system economically viable for smallholder farmers. This model is conservative: including the NEB carbon credit revenue at 10 USD/t CO<sub>2</sub>e further reduces the effective payback period to 2.7 yr for a 5-member cooperative (Section S9), consistent with the main-text analysis.

### S5.5 Tiered Subsidy Framework

We propose a tiered subsidy framework where the NEB serves as the verifiable metric for subsidy qualification:

- **Tier 1 (NEB < 5 kg CO<sub>2</sub>e/m<sup>2</sup>/yr):** 0 % subsidy
- **Tier 2 (NEB 5–10 kg CO<sub>2</sub>e/m<sup>2</sup>/yr):** 15 % subsidy
- **Tier 3 (NEB > 10 kg CO<sub>2</sub>e/m<sup>2</sup>/yr):** 30 % subsidy

This creates a direct incentive loop: higher quantum yield improvements lead to higher NEB, which unlocks greater subsidy access, enabling broader deployment.

## S6 BB84 Quantum Key Distribution: Implementation and Performance

### S6.1 Protocol Details

The BB84 protocol [?] is implemented as follows:

1. **Preparation:** Alice (sensor node) generates  $4N = 1024$  random bits (for  $N = 256$  final key length) and encodes each bit in one of four polarization states:  $|H\rangle$ ,  $|V\rangle$  (rectilinear basis,  $+$ ) or  $|D\rangle$ ,  $|A\rangle$  (diagonal basis,  $\times$ ).
2. **Transmission:** The quantum states are transmitted over a noisy classical channel with a simulated depolarizing noise rate of  $p = 5\%$ , modeling the decoherence in field-deployed fiber links [?].
3. **Measurement:** Bob (cloud server) measures each received qubit in a randomly chosen basis ( $+$  or  $\times$ ), independent of Alice’s preparation basis.
4. **Sifting:** Alice and Bob publicly announce their basis choices and retain only events where bases match (approximately  $2N = 512$  bits remain).
5. **Parameter estimation:** A random sample (minimum 1 bit, maximum 50 bits) of the sifted key is disclosed to estimate the quantum bit error rate (QBER). The sample fraction follows  $f_{\text{sample}} = \min(50/N, 0.2)$ .
6. **Key generation:** If  $\text{QBER} < 11\%$  (the security threshold), the remaining sifted bits form the raw key. Error correction and privacy amplification reduce the key to  $N = 256$  bits.

## S6.2 Error Threshold and Security

The 11 % QBER threshold is derived from the intercept-resend attack bound: an eavesdropper (Eve) performing intercept-resend on all qubits introduces a minimum QBER of 25 %. The 11 % threshold provides a safety margin for practical implementations [?, ?].

## S6.3 Fail-Safe Mechanism

If the QBER exceeds 11 % (e.g., due to extreme weather events, equipment malfunction, or attempted eavesdropping), the key generation aborts and a fail-safe local irrigation routine activates:

- A pre-programmed irrigation schedule (stored in non-volatile memory) ensures uninterrupted water supply to crops.
- The fail-safe schedule is based on the FAO-56  $\text{ET}_c$  estimate for the current season.
- Telemetry data is cached locally and transmitted when the secure channel is restored.

## S6.4 Computational Overhead

Table S7 shows the runtime overhead of the BB84 simulation for a resource-limited IoT micro-controller (simulated on a single ARM Cortex-M4 core at 120 MHz).

| Table S7: BB84 QKD simulation runtime overhead. |           |             |
|-------------------------------------------------|-----------|-------------|
| Operation                                       | Time (ms) | Memory (kB) |
| State preparation (1024 bits)                   | 0.8       | 0.5         |
| Noisy transmission                              | 1.2       | 0.1         |
| Measurement and sifting                         | 0.6       | 0.3         |
| QBER estimation                                 | 0.1       | 0.1         |
| Key distillation                                | 2.1       | 1.2         |
| Total                                           | 4.8       | 2.2         |

The total runtime of 4.8 ms and memory footprint of 2.2 kB are negligible for resource-limited IoT nodes, confirming the practical feasibility of BB84-secured agricultural telemetry.

## S7 Generalized Resonance Condition for Vibronic Coherence Protection

### S7.1 Theoretical Derivation

The resonance condition that enables the coherence protection mechanism is a general property of vibronically coupled chromophore aggregates [?, ?]. For a dimer of two chromophores with site energies  $\varepsilon_1, \varepsilon_2$  and electronic coupling  $J_{12}$ , the excitonic energy gap is:

$$\Delta E_{\text{exc}} = \sqrt{(\varepsilon_1 - \varepsilon_2)^2 + 4J_{12}^2}. \quad (\text{S11})$$

When this gap is resonant with an underdamped vibrational mode of frequency  $\omega_v$ :

$$\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{12}, \quad (\text{S12})$$

the excitonic and vibrational states hybridize to form a vibronic (polaron-dressed) state. The  $\pm J_{12}$  term accounts for Davydov splitting in the dimer: the symmetric and antisymmetric combinations of the two site excitations have energies separated by  $2J_{12}$ , each coupling differently to the vibrational manifold.

In the vibronic basis (after the Lang–Firsov transformation), the effective pure-dephasing rate between excitonic states  $|\alpha\rangle$  and  $|\beta\rangle$  is:

$$\Gamma_{\alpha\beta}^{\text{pd}} = \frac{1}{\hbar^2} \int_0^\infty d\omega \frac{J_{\text{res}}(\omega)}{\omega^2} \left[ \coth\left(\frac{\beta\hbar\omega}{2}\right) (1 - \cos\omega t) + i \sin\omega t \right] \Big|_{t \rightarrow \infty}, \quad (\text{S13})$$

where  $J_{\text{res}}(\omega)$  is the residual spectral density defined in Eq. S5. Modes satisfying Eq. S12 are removed from  $J_{\text{res}}(\omega)$ , reducing  $\Gamma_{\alpha\beta}^{\text{pd}}$  proportionally.

### S7.2 Generality Across Material Platforms

The resonance condition is not specific to the FMO complex. Any molecular aggregate satisfying the condition  $\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{nm}$  can benefit from spectral coherence protection. Relevant platforms include:

- **Synthetic J-aggregates:** Organic dyes that form excitonically coupled chains with tunable  $J_{nm}$  [?].
- **Plant light-harvesting complexes:** LHCII, CP29, and PSI core complexes that have underdamped vibrational modes in the 100–1 600  $\text{cm}^{-1}$  range.
- **Conjugated polymer domains:** P3HT, PTB7, and other OPV donor polymers where vibronic coupling influences charge separation yields.

### S7.3 Dual-Band Transmission Design Criterion

For the OPV spectral filter design, the transmission bands are chosen to satisfy the resonance condition for the dominant energy-transfer pathway in the target photosynthetic complex. The general design criterion is:

$$\lambda_{\text{filter}} \approx \frac{2\pi c}{\omega_{\text{ex}} \pm \omega_v}, \quad (\text{S14})$$

where  $\omega_{\text{ex}}$  is the primary exciton absorption frequency and  $\omega_v$  is the underdamped mode frequency. For FMO at 295 K,  $\omega_{\text{ex}} \approx 12\,500\text{ cm}^{-1}$  (800 nm) and  $\omega_v = 180\text{ cm}^{-1}$  gives  $\lambda_{\text{filter}} \approx 790\text{ nm}$  and  $810\text{ nm}$ , which defines the dual-band filter passband.

## S8 Three-Site Excitonic Trimer Verification

### S8.1 Motivation

To confirm that the dual-band filtering effect is not an artifact of the specific FMO complex parameters, we apply the protocol to a minimal three-site excitonic trimer with periodic boundary conditions [?, ?, ?].

### S8.2 Trimer Parameters

The trimer Hamiltonian is:

$$\hat{H}_{\text{trimer}} = \sum_{n=1}^3 \varepsilon_n |n\rangle\langle n| + J \sum_{n=1}^3 (|n\rangle\langle n+1| + |n+1\rangle\langle n|), \quad (\text{S15})$$

with  $\varepsilon_n = \varepsilon_0 + \delta_n$  (uniform site energy  $\varepsilon_0 = 12\,410\text{ cm}^{-1}$  plus small Gaussian disorder  $\delta_n \sim \mathcal{N}(0, 10\text{ cm}^{-1})$ ), and  $J = -50\text{ cm}^{-1}$ .

The spectral density is a single Drude–Lorentz mode ( $\lambda = 20\text{ cm}^{-1}$ ,  $\gamma = 30\text{ cm}^{-1}$ ) plus one underdamped mode at  $\omega_v = 200\text{ cm}^{-1}$  (Huang–Rhys factor  $S = 0.08$ ).

### S8.3 Results

Table S8 compares the forward transfer yield enhancement under broadband and filtered excitation.

Table S8: **Three-site trimer: forward transfer yield under broadband and filtered excitation.**

| Excitation           | $\Phi_{\text{FT}}$ | $\tau_c$ (fs) | $\eta_{\text{FT}}$ |
|----------------------|--------------------|---------------|--------------------|
| Broadband            | 0.62(4)            | 180(20)       | —                  |
| Filtered (dual-band) | 0.73(3)            | 240(25)       | 0.18(3)            |

The coherence enhancement  $\eta_{\text{FT}} = (\Phi_{\text{FT}}^{\text{flt}} - \Phi_{\text{FT}}^{\text{broad}})/\Phi_{\text{FT}}^{\text{broad}} = 0.18(3)$  is consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [?]. This demonstrates that the spectral filtering effect is a general vibronic phenomenon, not specific to the FMO complex.

## S9 Proof-of-Concept Scenario: 1 ha High-Tunnel Greenhouse in Cameroon

### S9.1 Site Description

We consider a 1 ha high-tunnel greenhouse located in Cameroon’s Western Highlands (latitude  $5.5^\circ\text{ N}$ , elevation  $1\,500\text{ m}$ ). This region has a mean annual solar irradiance of  $5.76\text{ kW h m}^{-2}\text{ d}^{-1}$  [?], a tropical highland climate with mean annual temperature  $22^\circ\text{ C}$ , and supports intensive tomato and vegetable production on smallholder farms that average  $0.5\text{ ha}$  to  $2\text{ ha}$  [?].

## S9.2 Water Savings

Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration ( $ET_c$ ) relative to the open-field FAO-56 baseline saves approximately  $13\,000\text{ m}^3\text{ yr}^{-1}$  of irrigation water for a 1 ha greenhouse. At the Cameroonian rural water consumption benchmark of  $50\text{ L d}^{-1}$  per household, this volume supplies 260 households annually.

## S9.3 Biomass Yield Improvement

The quantum yield enhancement ( $\eta_{\text{FT}} = 25\%$  under dual-band spectral filtering) translates directly to crop biomass via the excitonic yield multiplier. Under the NPoM shield (accounting for the plasmonic transport penalty, which reduces  $\Phi_{\text{FT}}$  from 0.98 to 0.08), the net biomass reaches 68 % of the open-field reference, compared to 40 % under conventional static shading. This represents an additional  $2.8\text{ t ha}^{-1}\text{ yr}^{-1}$  of tomato-equivalent fresh produce at local market prices.

## S9.4 Energy and Carbon Balance

The OPV module generates  $42\text{ MWh yr}^{-1}$  of electrical energy for a 1 ha installation at  $180\text{ kW h m}^{-2}\text{ yr}^{-1}$  generation rate. Displacing grid electricity at a carbon intensity of  $450\text{ g CO}_2\text{e/kWh}$  (Section S5) offsets  $19\text{ t CO}_2\text{e/yr}$ . Combined with the NEB from water savings and manufacturing footprint, the system achieves net-zero operational carbon in 3.2 yr (Figure 3 of the main text). At a conservative carbon credit price of  $10\text{ USD/t CO}_2\text{e}$  [?], the system generates approximately  $124\text{ USD yr}^{-1}$  in verified carbon revenue, which reduces the 5-member cooperative effective CAPEX payback to 2.7 yr.

## S9.5 Smallholder Applicability

Cameroon has approximately 10 000 smallholder farmers operating in the Western Highlands who could benefit from cooperative agrivoltaic deployment. The 1 ha module scale aligns with the typical landholding size (0.5 ha to 2 ha), and the 5-member cooperative model (Section S5) reduces individual capital exposure to approximately  $64\text{ m}^{-2}$  after subsidy. This scenario demonstrates that the quantum yield enhancement — despite the NPoM transport penalty — translates into measurable sustainability and economic benefits at the farm scale.